

FORESEE 嵌入式存储产品目录

中文工程翻译版

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翻译原则：型号、信号名、寄存器位、单位、公式、器件料号及标准编号保持原文；说明性英文译为中文。每个原文页面后附原页图示，便于核对曲线、封装图、时序图和复杂表格。

注意：本文件为工程参考翻译，不替代芯片/器件厂商发布的最新官方英文数据手册。设计定版时应以官方原始资料为准。

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嵌入式存储产品目录

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

Stock Code: 301308.SZ

FORESEE®
More than Foresee

EMBEDDED STORAGE PRODUCT CATALOGUE

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本版本更新日期为 2023 年 4 月 23 日。实际情况以江波龙（Longsys）官网发布的信息为准。

原页图示参考

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This version is to be updated on April 23, 2023. Actual conditions are subject to the Longsys website.

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【产品线介绍】

【Flash 系列产品】

eMMC：P.04-P.06

UFS：P.07

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SPI NAND Flash：P.09

【DRAM 系列产品】

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PRODUCT LINE INTRODUCTION

FLASH SERIES PRODUCTS

eMMC P.04-P.06

UFS P.07

Parallel NAND Flash P.08

SPI NAND Flash P.09

DRAM SERIES PRODUCTS

LPDDR P.10

DDR3L P.11

MULTI-CHIP PACKAGE SERIES PRODUCTS

eMCP P.12

uMCP P.13

ePOP P.14

NAND-based MCP P.15

BRAND INTRODUCTION

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【嵌入式存储产品线介绍】

FORESEE 嵌入式存储产品线成立于 2011 年。同年推出了中国首批实现量产的 eMMC 产品。经过 10 余年与行业客户共同成长，FORESEE 已陆续推出采用 Flash、DRAM 以及多芯片封装技术的嵌入式存储产品，广泛应用于汽车电子、工业控制、智能手机、平板电脑、智能电视、智能穿戴、网络通信等众多领域。FORESEE 还与 Qualcomm、MediaTek、UNISOC 等芯片平台厂商建立了长期、紧密的合作关系，为全球消费、工业及汽车领域企业提供稳定可靠的产品。

依托江波龙多项核心专利以及自主研发固件，FORESEE 嵌入式存储产品经过多项严格的验证测试，可全面满足系统流畅性和数据稳定性的要求。其中，车规级 eMMC 已通过 AEC-Q100 认证。AEC-Q100 是汽车电子委员会（AEC）制定的可靠性资格认证标准，也是汽车行业认可度较高的认证之一。

【技术特性】

FBA：利用空闲块的 pSLC 模式提升性能。

掉电保护（Power-loss Protection）：即使发生断电或电压不稳定，也能保证数据有效性。

预交换（Pre-exchange）：在错误发生前提前替换数据。

FFU：显著降低售后服务和维护的成本及难度。

主动整理（Actively Organizing）：SoC 平台可主动整理数据，从而提高读写性能。

增强模式（Enhance Mode）：对存储区域进行增强，使关键数据更加安全、可靠。

原页图示参考

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Embedded Storage Product Line Introduction

The FORESEE embedded storage product line was established in 2011. The first mass-produced eMMC in China was launched the same year. After over 10 years of mutual growth alongside industry clients, FORESEE has released embedded storage products which use Flash, DRAM, or multi-chip packaging. These types of products have become widely used in automotive electronics, industrial control, smartphones, tablets, smart TVs, smart wearables, network communication, and many other fields. FORESEE has also established long-term tight-knit cooperation with Qualcomm, MediaTek, UniSOC, and other chip platform suppliers to provide global enterprises in consumption, industrial, and automotive fields with stable and reliable products.

With Longsys's multiple core patents and in-house firmware, FORESEE embedded storage products have undergone multiple strict verification tests and can comprehensively meet the system smoothness and data stability requirements. Its automotive eMMC has passed AEC-Q100, the reliability standard qualification of the Automotive Electronics Council (AEC) and one of the most credible qualification for the automotive industry.

Technical Features



FBA

The pSLC mode of free blocks is used to boost performance.



Power-loss Protection

Power-loss Protection: Data validity is ensured even when power is turned off or voltage is unstable.



Pre-exchange

Data is replaced before errors occur.



FFU

The cost and difficulty of after-sales services and maintenance are greatly reduced.



Actively Organizing

The SoC platform can organize data actively to improve the read/write performance.



Enhance Mode

The storage area is enhanced to make key data more secure and reliable.

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【应用】
智能手机、平板电脑、智能电视、机顶盒、智能音箱、智能手表、智能摄像机、智能手环、智能耳机、车载多媒体娱乐、工业自动化、IPC、通信基站、Wi-Fi 路由器、商用显示设备、流媒体后视镜、行车记录仪等。

【车规级 eMMC Grade 2】												
表头：容量 料号 P/N 电压 存储介质 温度范围 接口 封装 包装 尺寸 状态												
4GB	FEMDME004G-A8A39	1.8V/3.3V	pSLC	-40 ° C~105 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品			
8GB	FEMDME008G-A8A39	1.8V/3.3V	MLC	-40 ° C~105 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品			
8GB	FEMDME008G-A8A43	1.8V/3.3V	pSLC	-40 ° C~105 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品			
16GB	FEMDME016G-A8A43	1.8V/3.3V	MLC	-40 ° C~105 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品			
16GB	FEMDME016G-A8A58	1.8V/3.3V	pSLC	-40 ° C~105 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品			
32GB	FEMDME032G-A8A58	1.8V/3.3V	MLC	-40 ° C~105 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品			
64GB	FEMDME064G-A8A58	1.8V/3.3V	MLC	-40 ° C~105 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品			

原页图示参考

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Automotive eMMC Grade2

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
4GB	FEMDME004G-A8A39	1.8V/3.3V	pSLC	-40°C to 105°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample
8GB	FEMDME008G-A8A39	1.8V/3.3V	MLC	-40°C to 105°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample
8GB	FEMDME008G-A8A43	1.8V/3.3V	pSLC	-40°C to 105°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample
16GB	FEMDME016G-A8A43	1.8V/3.3V	MLC	-40°C to 105°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample
16GB	FEMDME016G-A8A58	1.8V/3.3V	pSLC	-40°C to 105°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample
32GB	FEMDME032G-A8A58	1.8V/3.3V	MLC	-40°C to 105°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample
64GB	FEMDME064G-A8A58	1.8V/3.3V	MLC	-40°C to 105°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample

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【车规级 eMMC Grade 3】											
表头：容量 料号 P/N 电压 存储介质 温度范围 接口 封装 包装 尺寸 状态											
8GB	FEMDMW008G-88A39	1.8V/3.3V	MLC	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
16GB	FEMDMW016G-88A43	1.8V/3.3V	3D NAND	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
32GB	FEMDMW032G-88A19	1.8V/3.3V	3D NAND	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
64GB	FEMDMW064G-88A19	1.8V/3.3V	3D NAND	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
【工业宽温 eMMC】											
8GB	FEMDRW008G-88A39	1.8V/3.3V	MLC	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
16GB	FEMDRW016G-88A43	1.8V/3.3V	MLC	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
32GB	FEMDRW032G-88A19	1.8V/3.3V	3D NAND	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
64GB	FEMDRW064G-88A19	1.8V/3.3V	3D NAND	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
128GB	FEMDRW128G-88A19	1.8V/3.3V	3D NAND	-40 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
【工业级 eMMC】											
8GB	FEMDRM008G-A3A55	1.8V/3.3V	MLC	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	送样		
16GB	FEMDRM016G-A3A55	1.8V/3.3V	MLC	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	新品		
32GB	FEMDRM032G-A3A55	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	送样		
32GB	FSEIASLD-32G	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
64GB	FSEIASLD-64G	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
64GB	FEMDRM064G-A3A56	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
128GB	FEMDRM0128G-A3A56	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		
128GB	FSEIASLD-128G	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	量产		

原页图示参考

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Automotive eMMC Grade3

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
8GB	FEMDMW008G-88A39	1.8V/3.3V	MLC	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
16GB	FEMDMW016G-88A43	1.8V/3.3V	3D NAND	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
32GB	FEMDMW032G-88A19	1.8V/3.3V	3D NAND	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
64GB	FEMDMW064G-88A19	1.8V/3.3V	3D NAND	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production

Industrial Wide-temperature eMMC

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
8GB	FEMDRW008G-88A39	1.8V/3.3V	MLC	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
16GB	FEMDRW016G-88A43	1.8V/3.3V	MLC	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
32GB	FEMDRW032G-88A19	1.8V/3.3V	3D NAND	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
64GB	FEMDRW064G-88A19	1.8V/3.3V	3D NAND	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
128GB	FEMDRW128G-88A19	1.8V/3.3V	3D NAND	-40°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production

Industrial eMMC

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
8GB	FEMDRM008G-A3A55	1.8V/3.3V	MLC	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sampling
16GB	FEMDRM016G-A3A55	1.8V/3.3V	MLC	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	New
32GB	FEMDRM032G-A3A55	1.8V/3.3V	3D NAND	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sampling
32GB	FSEIASLD-32G	1.8V/3.3V	3D NAND	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
64GB	FSEIASLD-64G	1.8V/3.3V	3D NAND	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
64GB	FEMDRM064G-A3A56	1.8V/3.3V	3D NAND	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
128GB	FEMDRM128G-A3A56	1.8V/3.3V	3D NAND	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production
128GB	FSEIASLD-128G	1.8V/3.3V	3D NAND	-25°C to 85°C	eMMC 5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Production

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【商用 eMMC】										
表头：容量 料号 P/N 电压 存储介质 温度范围 接口 封装 包装 尺寸 状态										
8GB	FEMDNN008G-A3A55	1.8V/3.3V	3D TLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	11.5×13×0.8 mm	MP	(量产)
16GB	FEMDNN016G-A3A55	1.8V/3.3V	3D TLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	11.5×13×0.8 mm	MP	(量产)
32GB	FEMDNN032G-A3A55	1.8V/3.3V	3D TLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	11.5×13×0.8 mm	量产	
64GB	FEMDNN064G-A3A56	1.8V/3.3V	3D TLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	11.5×13×0.8 mm	量产	
128GB	FEMDNN128G-A3A56	1.8V/3.3V	3D TLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	11.5×13×0.8 mm	量产	
256GB	FEMDNN256G-A3A56	1.8V/3.3V	3D TLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	11.5×13×1.0 mm	样品	
【商用小尺寸 eMMC】										
4GB	FEMKNN004G-58A42	1.8V/3.3V	MLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	9×7.5×0.8 mm	量产	
8GB	FEMKNN008G-58A42	1.8V/3.3V	MLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	9×7.5×0.8 mm	样品	
32GB	FEMJNM032G-58C29	1.8V/3.3V	TLC	-25 °C~85 °C	eMMC 5.1	FBGA153	托盘	9×10×0.8 mm	量产	

【优势】

不同规格的 eMMC 可用于复杂且变化较大的应用场景，满足客户对高性能、低功耗、兼容性和稳定性的严格要求。

在符合 JEDEC 引脚标准的前提下，小尺寸 eMMC 具有更小的封装尺寸，可占用更少的 PCB 空间。

工业级、工业宽温级以及车规级 eMMC 符合 IATF 16949 标准，平均无故障时间（MTBF）超过 2000 万小时，具备高可靠性和长使用寿命，可支持设备长时间 7×24 小时持续运行。

车规级 eMMC 满足 AEC-Q100 可靠性验证标准。

原页图示参考

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Commercial eMMC

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
8GB	FEMDNN008G-A3A55	1.8V/3.3V	3D TLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	11.5 x 13 x 0.8 mm	MP
16GB	FEMDNN016G-A3A55	1.8V/3.3V	3D TLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	11.5 x 13 x 0.8 mm	MP
32GB	FEMDNN032G-A3A55	1.8V/3.3V	3D TLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	11.5 x 13 x 0.8 mm	Production
64GB	FEMDNN064G-A3A56	1.8V/3.3V	3D TLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	11.5 x 13 x 0.8 mm	Production
128GB	FEMDNN128G-A3A56	1.8V/3.3V	3D TLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	11.5 x 13 x 0.8 mm	Production
256GB	FEMDNN256G-A3A56	1.8V/3.3V	3D TLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	11.5 x 13 x 1.0 mm	Sample

Commercial Subsize eMMC

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
4GB	FEMKNN004G-58A42	1.8V/3.3V	MLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	9*7.5*0.8mm	Production
8GB	FEMKNN008G-58A42	1.8V/3.3V	MLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	9*7.5*0.8mm	Sample
32GB	FEMJNM032G-58C29	1.8V/3.3V	TLC	-25°C to 85°C	eMMC5.1	FBGA153	Tray	9*10*0.8mm	Production

Advantages

- eMMCs in various specifications can be used in complex and fluctuating application scenarios to meet clients' strict requirements for high performance, low power consumption, compatibility, and stability.
- Taking the JEDEC pin standards into account, the sub-size eMMC features a smaller size and less PCB space.
- The industrial/industrial wide-temperature/automotive eMMCs comply with the IATF 16949 standard. Boasting a mean time between failures (MTBF) of over 20 million hours, high reliability, and a long service life, they can ensure the around-the-clock operations of devices for extended periods.
- The automotive eMMCs meet the reliability validation standard AEC-Q100.

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【应用】
智能手机、智能平板、智能摄像机、VR/AR、智能驾驶汽车、多媒体娱乐、流媒体后视镜、行车记录仪等。

【车规级 UFS】									
表头：容量 料号 P/N 电压 存储介质 温度范围 接口 封装 包装 尺寸 状态									
64GB	FEUDME064G-B8A19	1.8V/3.3V	3D NAND	-40 ° C~105 ° C	UFS 2.1 Gear3 2L	FBGA153	托盘	11.5×13×1.2 mm	样品
128GB	FEUDME128G-B8A19	1.8V/3.3V	3D NAND	-40 ° C~105 ° C	UFS 2.1 Gear3 2L	FBGA153	托盘	11.5×13×1.2 mm	样品

【商用 UFS】									
64GB	FEUDNN064G-C2A56	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	UFS 2.2	FBGA153	托盘	11.5×13×1.0 mm	量产
64GB	FEUDNN064G-C2H14	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	UFS 2.2	FBGA153	托盘	11.5×13×1.0 mm	样品
128GB	FEUDNN128G-C2A56	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	UFS 2.2	FBGA153	托盘	11.5×13×1.0 mm	量产
128GB	FEUDNN128G-C2H14	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	UFS 2.2	FBGA153	托盘	11.5×13×1.0 mm	样品
256GB	FEUDNN256G-C2A44	1.8V/3.3V	3D NAND	-25 ° C~85 ° C	UFS 2.2	FBGA153	托盘	11.5×13×1.0 mm	量产

【优势】
大容量、高性能和更低功耗可帮助设备获得更长续航时间。
支持 UFS 2.2/UFS 3.1 的新特性，以及 LDPC 算法、Write Booster、HPB、紧急掉电保护、FFU 升级、FBA 数据加速等功能。

兼容主流平台。
车规级 UFS 满足 AEC-Q100 可靠性验证标准。
高可靠性和长寿命可支持设备长时间 7×24 小时持续运行。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



Automotive UFS

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
64GB	FEUDME064G-B8A19	1.8V/3.3V	3D NAND	-40 to 105	UFS 2.1 Gear3 2L	FBGA153	Tray	11.5 x 13 x 1.2mm	Sample
128GB	FEUDME128G-B8A19	1.8V/3.3V	3D NAND	-40 to 105	UFS 2.1 Gear3 2L	FBGA153	Tray	11.5 x 13 x 1.2mm	Sample

Commercial UFS

Density	P/N	Voltage	Memory	Temp. Range	Interface	Package	Packing	Size	Status
64GB	FEUDNN064G-C2A56	1.8V/3.3V	3D NAND	-25°C to 85°C	UFS2.2	FBGA153	Tray	11.5 x 13 x 1.0mm	Production
64GB	FEUDNN064G-C2H14	1.8V/3.3V	3D NAND	-25°C to 85°C	UFS2.2	FBGA153	Tray	11.5 x 13 x 1.0mm	Sample
128GB	FEUDNN128G-C2A56	1.8V/3.3V	3D NAND	-25°C to 85°C	UFS2.2	FBGA153	Tray	11.5 x 13 x 1.0mm	Production
128GB	FEUDNN128G-C2H14	1.8V/3.3V	3D NAND	-25°C to 85°C	UFS2.2	FBGA153	Tray	11.5 x 13 x 1.0mm	Sample
256GB	FEUDNN256G-C2A44	1.8V/3.3V	3D NAND	-25°C to 85°C	UFS2.2	FBGA153	Tray	11.5 x 13 x 1.0mm	Production

Advantages

- Large capacity, high performance and lower power consumption ensure longer battery life for devices
- New features of UFS 2.2/UFS 3.1, LDPC algorithm, Write Booster, HPB, emergency power failure protection, FFU upgrade, FBA data acceleration, and other functions are supported.
- Compatible with mainstream platforms
- The automotive UFS meet the reliability validation standard AEC-Q100.
- High reliability and long service life can ensure the around-the-clock operations of devices for extended periods.

原文第 9 页 - 中文译文

【应用】

GPON、机顶盒、IPC、智能音箱、无人机等。

【并行 NAND Flash（TSOP）】

表头：	容量	料号 P/N	电压	存储介质	温度范围	总线	封装	包装	尺寸
1Gb	FS33ND01GS108TFI0	3.3V	SLC	-40 ° C~85 ° C	x8	TSOP48	托盘	12×20×1.2 mm	
1Gb	FSNU8A001G-TWT	1.8V	SLC	-40 ° C~85 ° C	x8	TSOP48	托盘	12×20×1.2 mm	
1Gb	FSNS8A001G-TWT	3.3V	SLC	-40 ° C~85 ° C	x8	TSOP48	托盘	12×20×1.2 mm	
2Gb	FSNU8A002G-TWT	1.8V	SLC	-40 ° C~85 ° C	x8	TSOP48	托盘	12×20×1.2 mm	
2Gb	FSNS8A002G-TWT	3.3V	SLC	-40 ° C~85 ° C	x8	TSOP48	托盘	12×20×1.2 mm	
4Gb	FS33ND04GS108TFI0	3.3V	SLC	-40 ° C~85 ° C	x8	TSOP48	托盘	12×20×1.2 mm	

【并行 NAND Flash（BGA）】

1Gb	FSNS8A001G-EWT	3.3V	SLC	-40 ° C~85 ° C	x8	BGA48	托盘	8×6×1.0 mm
2Gb	FSNU8A002G-EWT	1.8V	SLC	-40 ° C~85 ° C	x8	BGA48	托盘	8×6×1.0 mm
4Gb	FS33ND04GS108BFI0	3.3V	SLC	-40 ° C~85 ° C	x8	BGA63	托盘	9×11×1.0 mm

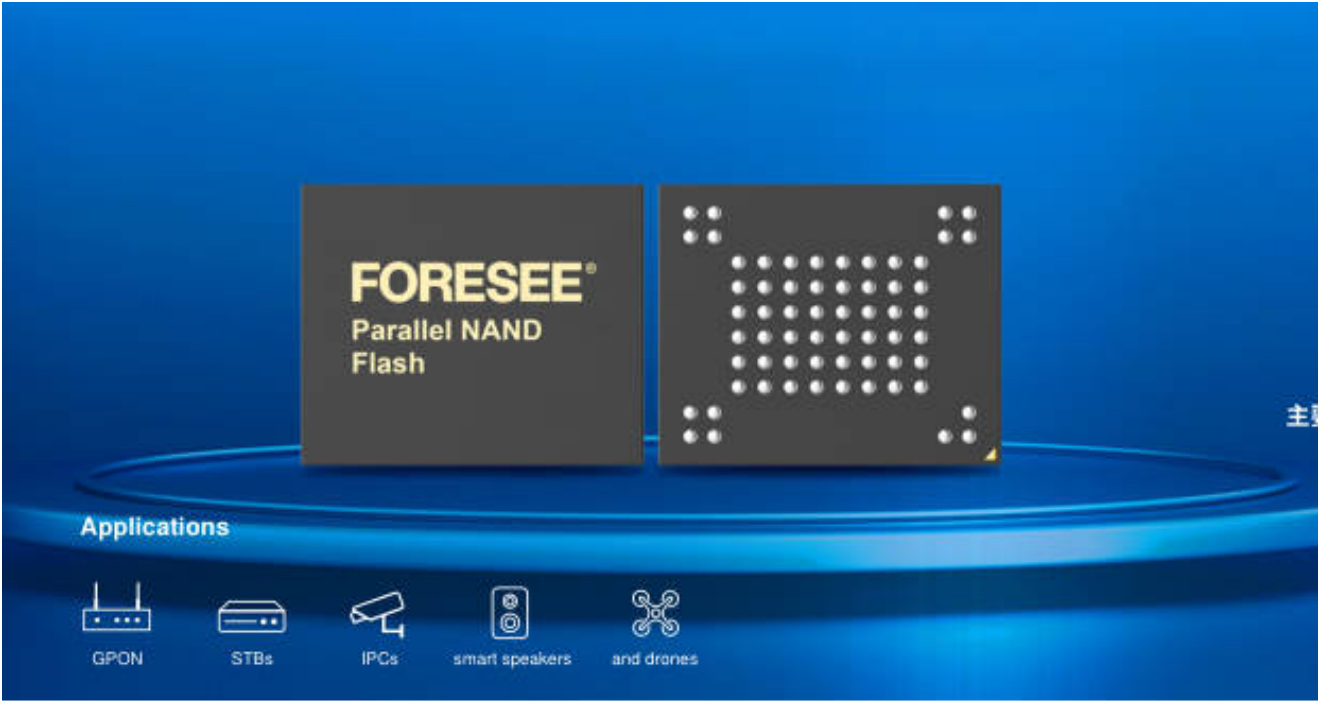
【优势】

主流产品支持 62×2K Bytes 的大容量 OTP 空间。

支持最小 48 Ball、8×6 mm 的 BGA 封装，可节省 PCB 空间。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



Parallel NAND Flash (TSOP)

Density	P/N	Voltage	Memory	Temp. Range	Bus	Package	Packing	Size
1Gb	FS33ND01GS108TFI0	3.3V	SLC	-40°C to 85°C	x8	TSOP48	Tray	12 × 20 × 1.2 mm
1Gb	FSNU8A001G-TWT	1.8V	SLC	-40°C to 85°C	x8	TSOP48	Tray	12 × 20 × 1.2 mm
1Gb	FSNS8A001G-TWT	3.3V	SLC	-40°C to 85°C	x8	TSOP48	Tray	12 × 20 × 1.2 mm
2Gb	FSNU8A002G-TWT	1.8V	SLC	-40°C to 85°C	x8	TSOP48	Tray	12 × 20 × 1.2 mm
2Gb	FSNS8A002G-TWT	3.3V	SLC	-40°C to 85°C	x8	TSOP48	Tray	12 × 20 × 1.2 mm
4Gb	FS33ND04GS108TFI0	3.3V	SLC	-40°C to 85°C	x8	TSOP48	Tray	12 × 20 × 1.2 mm

Parallel NAND Flash (BGA)

Density	P/N	Voltage	Memory	Temp. Range	Bus	Package	Packing	Size
1Gb	FSNS8A001G-EWT	3.3V	SLC	-40°C - 85°C	x8	BGA48	Tray	8 × 6 × 1.0 mm
2Gb	FSNU8A002G-EWT	1.8V	SLC	-40°C - 85°C	x8	BGA48	Tray	8 × 6 × 1.0 mm
4Gb	FS33ND04GS108BFI0	3.3V	SLC	-40°C to 85°C	x8	BGA63	Tray	9 × 11 × 1.0 mm

Advantages

- The mainstream products support 62x2K bytes large-capacity OTP space.
- A minimum BGA package of 48ball 8x6mm is supported to save PCB space.

原文第 10 页 - 中文译文

【应用】
GPON、Wi-Fi 路由器、智能音箱、IPC、机顶盒、可视门铃等。

【SPI NAND Flash (1.8 V)】									
表头：容量 料号 P/N 电压 存储介质 温度范围 总线 接口 封装 包装 尺寸									
512Mb	F35UQA512M-WWT	1.8V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	托盘	8×6×0.8 mm
512Mb	F35UQA512M-VWR	1.8V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	卷带	6×5×0.8 mm
1Gb	F35UQA001G-WWT	1.8V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	托盘	8×6×0.8 mm
1Gb	F35UQA001G-VWR	1.8V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	卷带	6×5×0.8 mm
2Gb	F35UQA002G-WWT	1.8V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	托盘	8×6×0.8 mm

【SPI NAND Flash (3.3 V)】									
512Mb	F35SQA512M-WWT	3.3V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	托盘	8×6×0.8 mm
512Mb	F35SQA512M-VWR	3.3V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	卷带	6×5×0.8 mm
1Gb	F35SQA001G-WWT	3.3V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	托盘	8×6×0.8 mm
1Gb	F35SQA001G-VWR	3.3V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	卷带	6×5×0.8 mm
2Gb	F35SQA002G-WWT	3.3V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	托盘	8×6×0.8 mm
4Gb	FS35ND04G-S2Y2QWFI000	3.3V	SLC	-40 ° C~85 ° C	x1/x2/x4	SPI	WSON8	托盘	8×6×0.8 mm

【优势】
产品采用通用 SPI 接口，并集成内部纠错码（ECC），可提高数据传输效率并保证数据可靠性。
支持最小 6×5 mm WSON 封装，可节省 PCB 空间。
作为中国大陆首款 512Mb SPI NAND Flash 产品之一，该产品采用优化晶圆设计，可帮助客户降低整机系统成本，同时提升产品竞争力。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



SPI NAND Flash (1.8V)

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size
512Mb	F35UQA512M-WWT	1.8V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	Tray	8 x 6 x 0.8 mm
512Mb	F35UQA512M-VWR	1.8V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	T&R	6 x 5 x 0.8 mm
1Gb	F35UQA001G-WWT	1.8V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	Tray	8 x 6 x 0.8 mm
1Gb	F35UQA001G-VWR	1.8V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	T&R	6 x 5 x 0.8 mm
2Gb	F35UQA002G-WWT	1.8V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	Tray	8 x 6 x 0.8 mm

SPI NAND Flash (3.3V)

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size
512Mb	F35SQA512M-WWT	3.3V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	Tray	8 x 6 x 0.8 mm
512Mb	F35SQA512M-VWR	3.3V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	T&R	6 x 5 x 0.8 mm
1Gb	F35SQA001G-WWT	3.3V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	Tray	8 x 6 x 0.8 mm
1Gb	F35SQA001G-VWR	3.3V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	T&R	6 x 5 x 0.8 mm
2Gb	F35SQA002G-WWT	3.3V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	Tray	8 x 6 x 0.8 mm
4Gb	F535ND04G-S2Y2QWF1000	3.3V	SLC	-40°C to 85°C	x1/x2/x4	SPI	WSON8	Tray	8 x 6 x 0.8 mm

Advantages

- The product is equipped with a general-purpose SPI and internal error correcting code (ECC) to improve data transmission efficiency and ensure data reliability.
- A minimum WSON 6mm*5mm package is supported to save PCB space.
- As the first 512Mb SPI NAND Flash product in mainland China, it comes with optimized wafer design, helping clients reduce the costs of complete machine systems while also making their products more competitive.

原文第 11 页 - 中文译文

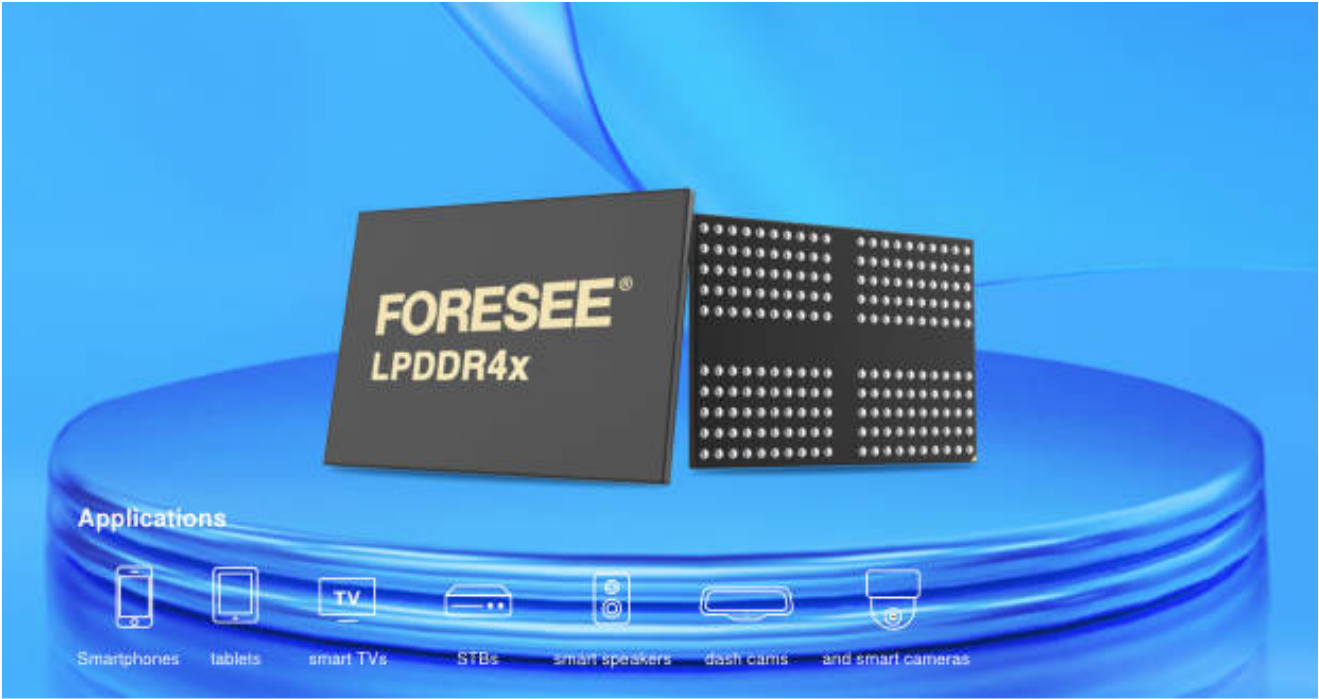
【应用】
智能手机、平板电脑、智能电视、机顶盒、智能音箱、行车记录仪、智能摄像机等。

【LPDDR4x】											
表头：容量 料号 P/N 电压 工艺/存储 温度范围 总线 接口 封装 包装 尺寸 状态											
8Gb	NCLDXC1MG256M32	1.8V/1.1V/1.1V 或 0.6V	1ynm	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
16Gb	FLXC2002G-N2	1.8V/1.1V/1.1V 或 0.6V	1α nm（原文字符异常）	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
16Gb	FLXC2002G-C4	1.8V/1.1V/0.6V	1ynm	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
24Gb	FLXC2003G-49	1.8V/1.1V/1.1V 或 0.6V	1znm	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
24Gb	FLXC2003G-C7	1.8V/1.1V/0.6V	1znm	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
24Gb	FLXC4003G-50	1.8V/1.1V/1.1V 或 0.6V	1xnm	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
32Gb	FLXC2004G-30	1.8V/1.1V/1.1V 或 0.6V	1α nm（原文字符异常）	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
32Gb	FLXC2004G-N1	1.8V/1.1V/1.1V 或 0.6V	1α nm（原文字符异常）	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
48Gb	FLXC4006G-20	1.8V/1.1V/1.1V 或 0.6V	1znm	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	量产	
64Gb	FLXC4008G-30	1.8V/1.1V/1.1V 或 0.6V	1α nm（原文字符异常）	-25 ° C~85 ° C	x32	LPDDR4x	FBGA200	托盘	10×14.5×1.0 mm	新品	

【优势】
RAM 产品广泛应用于便携式设备。
可提升系统性能并降低系统功耗。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



LPDDR4x

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size	Status
8Gb	NCLDXC1MG256M32	1.8V/1.1V/1.1V or 0.6V	1ynm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
16Gb	FLXC2002G-N2	1.8V/1.1V/1.1V or 0.6V	10nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
16Gb	FLXC2002G-C4	1.8V/1.1V/0.6V	1ynm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
24Gb	FLXC2003G-49	1.8V/1.1V/1.1V or 0.6V	12nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
24Gb	FLXC2003G-C7	1.8V/1.1V/ 0.6V	12nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
24Gb	FLXC4003G-50	1.8V/1.1V/1.1V or 0.6V	12nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
32Gb	FLXC2004G-30	1.8V/1.1V/1.1V or 0.6V	10nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
32Gb	FLXC2004G-N1	1.8V/1.1V/1.1V or 0.6V	10nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
48Gb	FLXC4006G-20	1.8V/1.1V/1.1V or 0.6V	12nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	Production
64Gb	FLXC4008G-30	1.8V/1.1V/1.1V or 0.6V	10nm	-25°C to 85°C	x32	LPDDR4x	FBGA200	Tray	10*14.5*1.0mm	New

Advantages

- RAM products that are widely used for portable devices
- Improves performance and reduces system power consumption.

原文第 12 页 - 中文译文

【应用】
无人机、家用 IPC、工业 IPC、机顶盒、智能音箱、智能电视、GPON、POS 机、电视盒等。

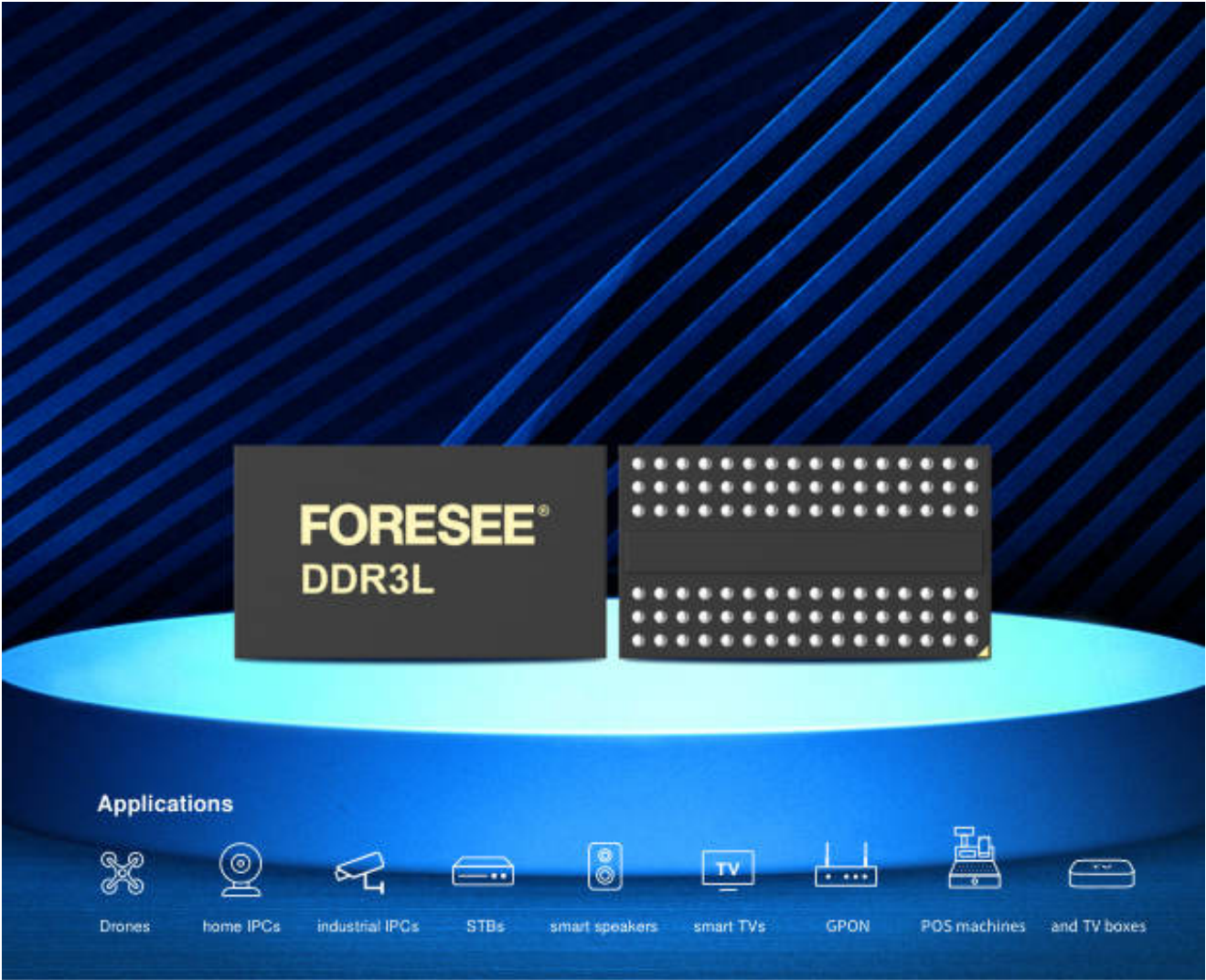
【DDR3L】

表头：	容量	料号 P/N	电压	存储介质	温度范围	总线	接口	封装	包装	尺寸
	2Gb	F60C1A0002-M6AR	1.35 或 1.5V	DDR3L	0 ° C~95 ° C	x16	DDR3	FBGA96	托盘	7.5×13.5×1.2 mm
	2Gb	F60C1A0002-M69W	1.35 或 1.5V	DDR3L	-40 ° C~95 ° C	x16	DDR3	FBGA96	托盘	7.5×13.5×1.2 mm
	4Gb	F60C1A0004-M79R	1.35 或 1.5V	DDR3L	0 ° C~95 ° C	x16	DDR3	FBGA96	托盘	7.5×13.5×1.2 mm
	4Gb	F60C1A0004-M79W	1.35 或 1.5V	DDR3L	-40 ° C~95 ° C	x16	DDR3	FBGA96	托盘	7.5×13.5×1.2 mm

【优势】
采用 25 nm 工艺并符合 JEDEC 标准。
产品可满足网络通信、消费电子和智能设备的数据缓存需求。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



DDR3L

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size
2Gb	F60C1A0002-M6AR	1.35 or 1.5V	DDR3L	0°C to 95°C	x16	DDR3	FBGA96	Tray	7.5 × 13.5 × 1.2 mm
2Gb	F60C1A0002-M69W	1.35 or 1.5V	DDR3L	-40°C to 95°C	x16	DDR3	FBGA96	Tray	7.5 × 13.5 × 1.2 mm
4Gb	F60C1A0004-M79R	1.35 or 1.5V	DDR3L	0°C to 95°C	x16	DDR3	FBGA96	Tray	7.5 × 13.5 × 1.2 mm
4Gb	F60C1A0004-M79W	1.35 or 1.5V	DDR3L	-40°C to 95°C	x16	DDR3	FBGA96	Tray	7.5 × 13.5 × 1.2 mm

Advantagess

- It adopts the 25nm process and complies with JEDEC standards.
- The product meets the data caching demands of network communication, consumer electronics, and smart devices.

原文第 13 页 - 中文译文

【应用】

智能手机、智能平板、智能穿戴、智能音箱、儿童手表、行车记录仪、商用显示设备、智能电视、机顶盒、Wi-Fi 路由器等。

【eMCP3】

表头：容量 料号 P/N 电压 存储组合 温度范围 总线 接口 封装 包装 尺寸 状态											
16GB+8Gb	FEPNA1608-58A4302	1.8V/3.3V	MLC+20nm	-25 ° C~85 ° C	x8+x32	eMCP3	FBGA221	托盘	11.5×13×1.0 mm	量产	
16GB+8Gb	FEPNA1608-58A4324	1.8V/3.3V	MLC+20nm	-25 ° C~85 ° C	x8+x32	eMCP3	FBGA221	托盘	11.5×13×1.0 mm	量产	
32GB+8Gb	FEPNA3208-58A1924	1.8V/3.3V	3D NAND+20nm	-25 ° C~85 ° C	x8+x32	eMCP3	FBGA221	托盘	11.5×13×1.0 mm	量产	

【eMCP4x】

64GB+32Gb	FEPRF6432-58A1930	eMMC:1.8V/3.3V ; LPDDR:1.8V/1.1V/1.1V 或 0.6V 3D NAND+1znm -25 ° C~85 ° C x8+x32 eMCP4x FBGA254 托盘 11.5×13×1.0 mm 量产									
64GB+32Gb	FEPRB6432-A3A56N1	eMMC:1.8V/3.3V ; LPDDR:1.8V/1.1V/1.1V 或 0.6V 3D NAND+1anm -25 ° C~85 ° C x8+x32 eMCP4x FBGA254 托盘 11.5×13×1.0 mm 量产									
128GB+32Gb	FEPRFA832-A3A56N1	eMMC:1.8V/3.3V ; LPDDR:1.8V/1.1V/1.1V 或 0.6V 3D NAND+1anm -25 ° C~85 ° C x8+x32 eMCP4x FBGA254 托盘 11.5×13×1.0 mm 量产									

【优势】

将 eMMC 与 LPDDR 组合在一个封装中。
节省 PCB 空间，有利于实现更紧凑、更轻量的智能设备。
eMMC 和 LPDDR 通过高精度集成封装，可提供 ECC、CRC、掉电保护、固件备份、磨损均衡等功能。
可有效帮助客户提高使用效率、降低使用成本、改善存储体验，并增强产品的连续性和稳定性。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



eMCP3

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size	Status
16GB+8Gb	FEPNA1608-58A4302	1.8V/3.3V	MLC+20nm	-25°C to 85°C	x8+ x32	eMCP3	FBGA221	Tray	11.5 x 13 x 1.0 mm	Production
16GB+8Gb	FEPNA1608-58A4324	1.8V/3.3V	MLC+20nm	-25°C to 85°C	x8+ x32	eMCP3	FBGA221	Tray	11.5 x 13 x 1.0 mm	Production
32GB+8Gb	FEPNA3208-58A1924	1.8V/3.3V	3D NAND+20nm	-25°C to 85°C	x8+ x32	eMCP3	FBGA221	Tray	11.5 x 13 x 1.0 mm	Production

eMCP4x

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size	Status
64GB+32Gb	FEPRF6432-58A1930	eMMC:1.8V/3.3V LPDDR: 1.8V/1.1/1.1or 0.6	3D NAND+12nm	-25°C to 85°C	x8+ x32	eMCP4x	FBGA254	Tray	11.5 x 13 x 1.0 mm	Production
64GB+32Gb	FEPRB6432-A3A56N1	eMMC:1.8V/3.3V LPDDR: 1.8V/1.1/1.1or 0.6	3D NAND+16nm	-25°C to 85°C	x8+ x32	eMCP4x	FBGA254	Tray	11.5 x 13 x 1.0 mm	Production
128GB+32Gb	FEPRFA832-A3A56N1	eMMC:1.8V/3.3V LPDDR: 1.8V/1.1/1.1or 0.6	3D NAND+16nm	-25°C to 85°C	x8+ x32	eMCP4x	FBGA254	Tray	11.5 x 13 x 1.0 mm	Production

Advantagess

- Combines eMMC and LPDDR
- Saves PCB space, enabling compact and lightweight smart devices.
- eMMC and LPDDR are integrated and packaged with precision to provide unique ECC, CRC, power failure protection, firmware backup, balanced wear, and more.
- It can effectively help clients boost usage efficiency, reduce usage costs, improve storage experience, and enhance product continuity and stability.

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【应用】
智能手机、智能平板、高端智能电视、高端商用显示设备、VR/AR、智能摄像机等。

【uMCP】
表头：容量 | 料号 P/N | 电压 | 存储组合 | 温度范围 | 总线 | 接口 | 封装 | 包装 | 尺寸
64GB+32Gb | FUPRB6432-C2A56N1 | UFS:1.8V/3.3V ; LPDDR:1.8V/1.1V/1.1V 或 0.6V | 3D NAND+1nm | -25 ° C~85 ° C | x8+x32 | uMCP4x | FBGA254 | 托盘 | 11.5×13×1.0 mm
128GB+32Gb | FUPRFA832-C2A56N1 | UFS:1.8V/3.3V ; LPDDR:1.8V/1.1V/1.1V 或 0.6V | 3D NAND+1nm | -25 ° C~85 ° C | x8+x32 | uMCP4x | FBGA254 | 托盘 | 11.5×13×1.0 mm

【优势】
将 LPDDR 与 UFS 组合，实现高性能和大容量。与分立式存储方案相比，可减少约 40% 的 PCB 占用面积。
减少存储芯片所需的 PCB 空间，使系统设计更加灵活。
可为智能手机和智能设备设计提供高密度、低功耗的存储解决方案。

原页图示参考

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Applications

 Smartphones

 smart tablets

 high-end smart TVs

 high-end commercial display devices

 VR/AR

 and smart cameras

uMCP

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size
64GB+ 32Gb	FUPRB6432-C2A56N1	UFS: 1.8V/3.3V LPDDR: 1.8V/1.1/1.1or 0.6	3D NAND+ 1nm	-25°C to 85°C	x8+ x32	uMCP4x	FBGA254	Tray	11.5 x 13 x 1.0 mm
128GB+ 32Gb	FUPRFA832-C2A56N1	UFS: 1.8V/3.3V LPDDR: 1.8V/1.1/1.1or 0.6	3D NAND+ 1nm	-25°C to 85°C	x8+ x32	uMCP4x	FBGA254	Tray	11.5 x 13 x 1.0 mm

Advantagess

- LPDDR and UFS are combined to enable high performance and large capacity. Compared with distributed storage solutions, it takes up 40% less PCB space.
- The PCB space of memory chips is reduced to enable more flexible system designs.
- Enabling high-density and low power consumption storage solutions for smartphone and smart device designs

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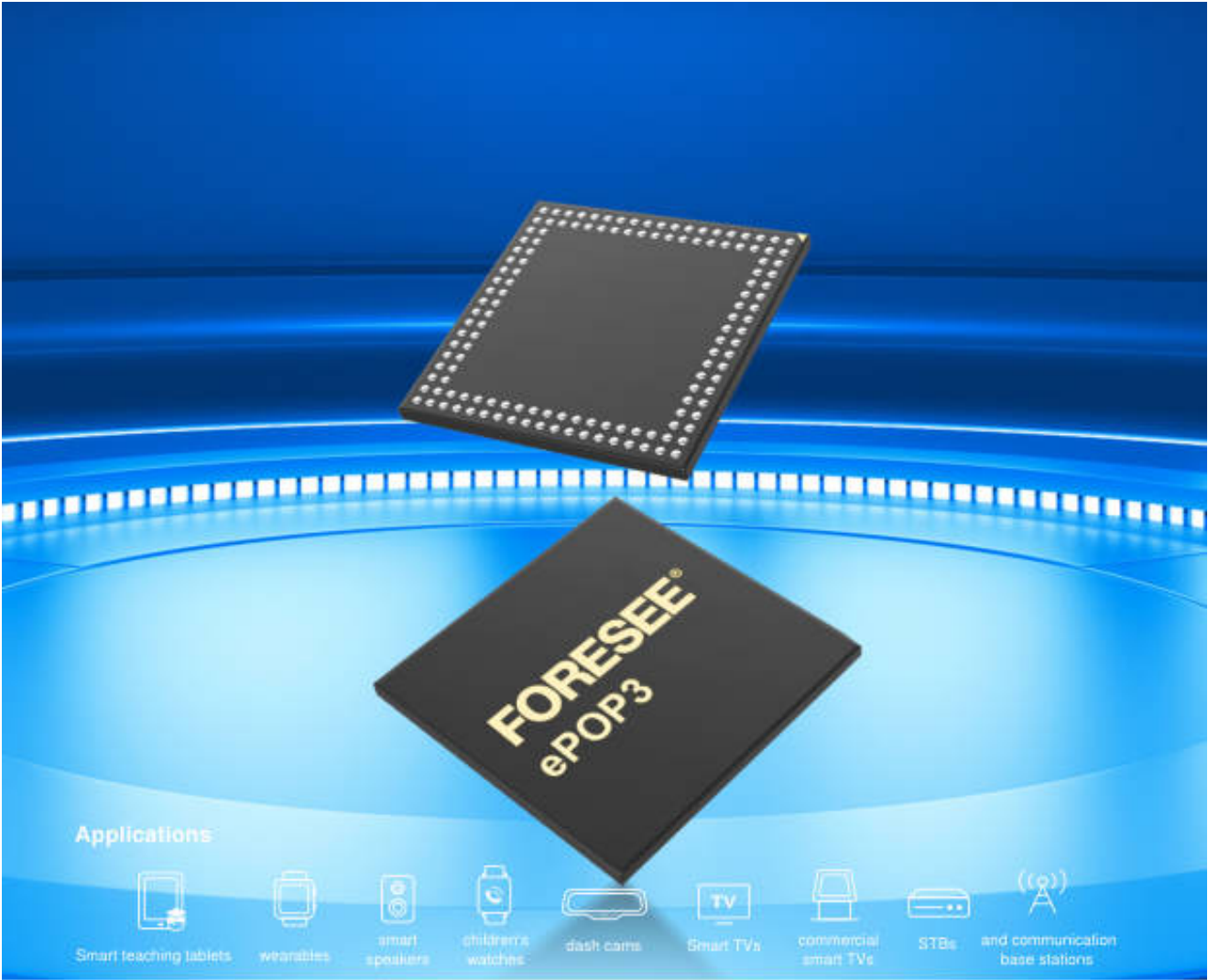
【应用】
智慧教育平板、智能穿戴、智能音箱、儿童手表、行车记录仪、智能电视、商用智能电视、机顶盒、通信基站等。

【ePOP3】											
表头：容量 料号 P/N 电压 存储组合 温度范围 总线 接口 封装 包装 尺寸											
8GB+4Gb	FAPUA0804-58C2948	1.8V/3.3V+1.2V	3D NAND+20nm	-25 ° C~85 ° C	x8+x32	ePOP3	FBGA136	托盘	10×10×0.9 mm		
8GB+8Gb	FAPUB0808-58C2948	1.8V/3.3V+1.2V	3D NAND+20nm	-25 ° C~85 ° C	x8+x32	ePOP3	FBGA136	托盘	10×10×0.9 mm		
32GB+8Gb	FAPUB3208-58C2948	1.8V/3.3V+1.2V	3D NAND+20nm	-25 ° C~85 ° C	x8+x32	ePOP3	FBGA136	托盘	10×10×0.9 mm		
【ePOP4x】											
32GB+16Gb	FAPEA3216-58C29N3	1.8V/3.3V+0.6V	3D NAND+1anm	-25 ° C~85 ° C	x8+x16	ePOP4x	FBGA144	托盘	8×9.5×0.8 mm		

【优势】
将 eMMC 与 LPDDR 集成为复合封装，专门面向可穿戴设备设计。
与 eMCP 产品相比，其厚度更薄、尺寸更小、功耗更低。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



ePOP3

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size
8GB+4Gb	FAPUA0804-58C2948	1.8V/3.3V+1.2V	3D NAND+20nm	-25°C to 85°C	x8+ x32	ePOP3	FBGA136	Tray	10*10*0.9mm
8GB+8Gb	FAPUB0808-58C2948	1.8V/3.3V+1.2V	3D NAND+20nm	-25°C to 85°C	x8+ x32	ePOP3	FBGA136	Tray	10*10*0.9mm
32GB+8Gb	FAPUB3208-58C2948	1.8V/3.3V+1.2V	3D NAND+20nm	-25°C to 85°C	x8+ x32	ePOP3	FBGA136	Tray	10*10*0.9mm

ePOP4x

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size
32GB+16Gb	FAPEA3216-58C29N3	1.8V/3.3V+0.6V	3D NAND+1nm	-25°C to 85°C	x8+ x16	ePOP4x	FBGA144	Tray	8*9.5*0.8mm

Advantagess

- Integrating eMMC and LPDDR, this compound product is tailor-made for wearables.
- Compared with eMCP products, it is thinner, smaller, and consumes less energy.

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【应用】
MiFi、POS 机、功能手机、4G 模组、儿童手表等。

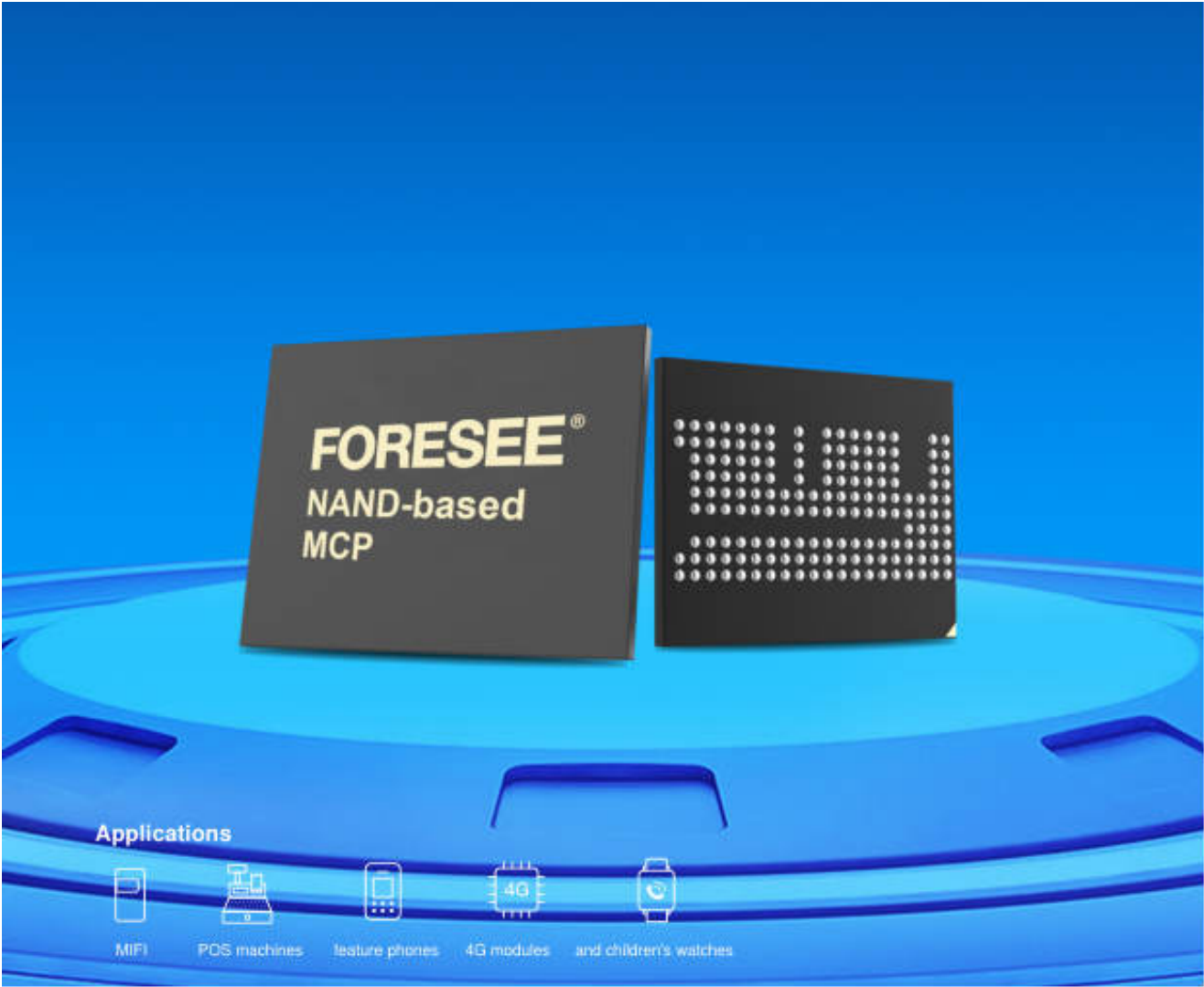
【nMCP】
表头：容量 | 料号 P/N | 电压 | 存储组合 | 温度范围 | 总线 | 接口 | 封装 | 包装 | 尺寸 | 状态
1Gb+1Gb | F70ME0101D-RDWA | 1.8V | SLC+LP2 | -40 ° C~85 ° C | x8+x32 | LPDDR2 | BGA162 | 托盘 | 8×10.5×1.0 mm | 量产

【优势】
Flash 与 LPDDR 封装在同一个器件中，可简化布线设计并节省空间。
1.8V 核心电压可满足可穿戴设备和物联网（IoT）对低功耗的要求。

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nMCP

Density	P/N	Voltage	Memory	Temp. Range	Bus	Interface	Package	Packing	Size	Status
1Gb+1Gb	F70ME01010-RDWA	1.8V	SLC+LP2	-40°C to 85°C	x8 + x32	LPDDR2	BGA162	Tray	6 x 10.5 x 1.0mm	Production

优势

- Flash and LPDDR are packaged in a single package to simplify wiring design and save space.
- A core voltage of 1.8V can meet the low power consumption requirements of wearable devices and the Internet of Things (IoT).

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原文第 17 页 - 中文译文

【关于 FORESEE】

FORESEE 创立于 2011 年，现已发展为专注于工业应用的存储品牌。FORESEE 以技术为驱动力、以客户为中心、以创新为导向，多年来持续深耕存储行业，目前拥有四大产品线：嵌入式存储（包括工业存储）、移动存储、固态硬盘和内存模组。

FORESEE 持续投入自主研发并建立全面质量控制体系，同时根据行业发展趋势和市场需求开发新产品，以优化性能、扩大存储容量、降低功耗并提升整体质量。FORESEE 还提供专业、及时的技术支持和市场服务，可提供工业级存储产品解决方案，并依托丰富的交付经验，为客户提供高效、创新且高品质的存储产品与解决方案。此外，FORESEE 已与多个行业的高端客户建立长期、稳定、深入的合作关系。

作为一个具有行业价值导向和高度责任感的品牌，FORESEE 将持续提升在存储行业的竞争力，以前沿、高品质技术和优质服务赋能各类应用创新、未来存储发展以及整个产业生态。

原页图示参考

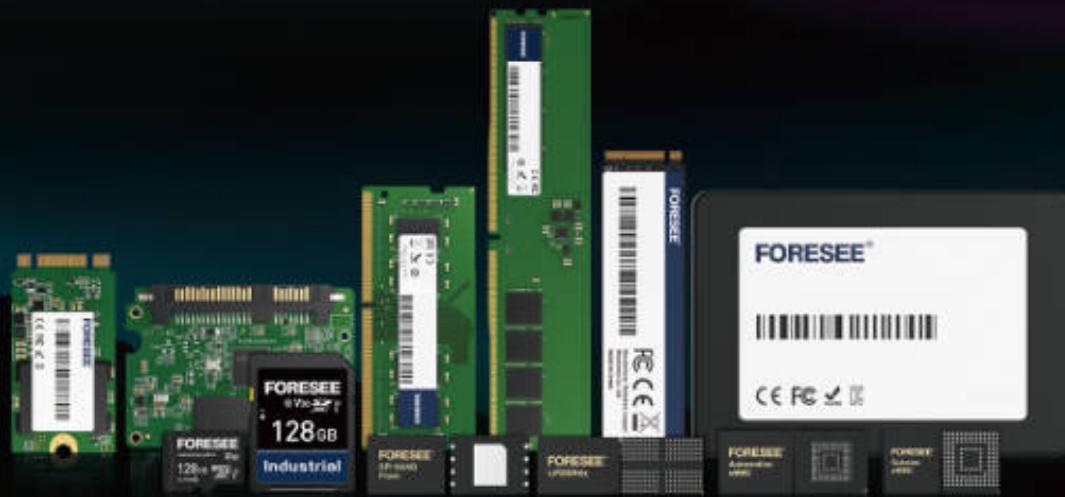
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About FORESEE

Founded in 2011, FORESEE has burgeoned into a storage brand specializing in industrial applications. Powered by technology, focused on customers, and driven by innovation, FORESEE has been engaged in the storage industry for many years. It owns four product lines: embedded storage (including industrial storage), mobile memory, solid-state drive, and memory module.

FORESEE has made continuous efforts in independent R&D as well as comprehensive quality control. It also develops new products based on industry dynamics and market needs to optimize performance, expand storage capacity, reduce power consumption, and improve overall quality. FORESEE also provides professional and instantaneous technical support and market services. FORESEE is capable of providing industrial-grade storage product solutions, also prides itself in taking advantage of its ample delivery experience to provide customers with effective, innovative, and quality storage products and solutions. In addition, FORESEE has established long-term, stable, and in-depth cooperative relationships with high-end customers in many industries.

As a brand with industry-oriented values and a strong sense of responsibility, FORESEE will continue to bolster its competitiveness edge within the storage industry. FORESEE will offer cutting-edge and high-quality technologies and superior services to empower not only the various application innovation and future storage development fields, but also the entire industry.



原文第 18 页 - 中文译文

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